

From Gentlemanly Status to Professional Breadwinner: Structural Topic Modeling of Masculine Formations in English-Language Fiction, 1771–1930

[Author(s) redacted for review]

Abstract

Gender theory holds that masculinity is plural and historically dynamic, yet measuring that plurality across a literary corpus invites two methodological hazards: fixing the number of topics so that it reproduces one’s own categories (circularity), and reading false precision from textual segments that are not independent. We confront both in a structural topic model (STM) of 150 English-language (British and American) novels (1771–1930), divided into 9,271 segments and modeled with period and author-gender metadata. Seven theory-driven masculine families serve as an interpretive lens rather than a target: model size is fixed on a substantive criterion—the smallest number of topics at which a professional-commercial formation becomes distinct ($K = 7$)—and uncertainty is estimated by bootstrapping whole novels rather than segments. Six of the seven families separate as distinct topics; the seventh, Christian moralism, does not separate until $K = 9$, dispersed across domestic and historical-religious discourse. Bootstrapped between-period differences are statistically robust for every formation: professional-commercial masculinity rises and a sentimental register declines most strongly, while the much-discussed “decline of the gentleman,” though robust, is roughly five times smaller in magnitude—so the shift is better described as a large new formation emerging *alongside* a modest erosion of the old than as a symmetric replacement. These trajectories converge with established literary historiography, corroborating the formations without proving them. We conclude that topic modeling can evidence the *existence* and *historical reweighting* of plural masculinities, but cannot certify their *number*—which remains a question for theory and interpretation rather than for the model.

Keywords: computational literary studies; digital humanities; structural topic modeling; masculinity; nineteenth-century fiction; gender history; distant reading; uncertainty quantification

1 Introduction

That masculinity is plural is now a settled premise of gender studies. Since Connell [5], scholars have treated manhood not as a single essence but as a field of competing arrangements—hegemonic and marginal, residual and emergent—organized through class, work, domesticity, religion, and empire. For the long nineteenth century, historians and literary critics have shown in detail how a gentleman, a father, a soldier, a clerk, and a man of God inhabit different and often incompatible scripts of manliness [1, 7, 13, 21, 23]. What this scholarship establishes through close reading, however, it rarely measures at scale: how the *relative weight* of these scripts shifts across a century of fiction, and whether such shifts can be observed reproducibly rather than inferred from a handful of canonical examples.

Computational literary studies promises exactly this kind of aggregate evidence [9, 11, 15, 17, 24]. Topic modeling, in particular, infers recurrent thematic structure without predefining every category,

and structural topic modeling (STM) lets document metadata such as period and authorship inform that structure [19]. But applying these tools to a theoretical construct like plural masculinity raises two hazards that are easy to overlook. The first is *circularity*: if one fixes the number of topics so that it reproduces a pre-chosen number of categories, the finding that “the model recovers the categories” is guaranteed by construction. The second is *false precision*: novels are typically split into many segments to capture local variation, but segments from the same book are not independent, so significance tests that treat them as such report implausibly tiny p -values. Both hazards push toward confident conclusions that the data do not support.

This article treats those two hazards as its central methodological problem rather than as footnotes. We model masculinity not as a list of male-coded words but as a set of recurrent thematic *formations*—constellations of co-occurring vocabulary through which masculine roles become socially legible. Seven theory-driven families (gentlemanly status, domestic paternalism, aristocratic chivalry, professional breadwinning, imperial adventure, Christian moralism, and sentimental manhood) serve as an interpretive lens, not as a target to be matched. Three design choices follow from this stance: we fix the model size on a substantive criterion independent of the taxonomy, estimate uncertainty by resampling whole novels, and validate the resulting formations against independent literary historiography. A topic model never “discovers” masculinity in the sense in which a critic interprets a scene; it produces artifacts that become evidence only when read cautiously [3].

The analysis yields three findings. First, six of the seven theoretical formations separate as distinct topics, and their *differential* separability is itself informative: Christian moralism does not emerge as its own topic until a higher model size, because in this corpus religious manliness is expressed *through* domestic discourse rather than apart from it. Second, the corpus exhibits a robust rise of a professional-commercial formation and a robust decline of a sentimental register; the celebrated “decline of the gentleman,” though also statistically robust, is roughly five times smaller in magnitude than the breadwinner’s rise—a mild corrective to a strand of the qualitative literature that treats the two movements as a symmetric exchange. Third, and most generally, topic modeling can support claims about the existence and historical reweighting of plural masculine formations while remaining unable to certify how many such formations exist, because the number of topics is a modeling choice rather than a quantity the data determine.

2 Background

Topic modeling entered literary history as a way to infer themes without hand-coding them, whether to mine the “great unread” for passages of interest [22] or to chart themes at corpus scale. Jockers and Mimno [12], the closest precedent to this study, used topics as proxies for literary themes and related their prevalence to author gender, nationality, and date. We depart from that work in two respects: we adopt STM rather than latent Dirichlet allocation [2] so that period and authorship inform topical prevalence directly [18, 19], and we narrow the focus to the historical movement of masculine formations specifically. A complementary strand models gender directly rather than thematically: Underwood et al. [25] show that the linguistic marking of gendered characters weakens across English-language fiction since the nineteenth century even as the share of fiction written by women declines. We instead track masculine *thematic* formations and their changing prevalence, a different and narrower object. The interpretive stakes of the method are themselves contested. Buurma [3] reads topic models as fictions that can usefully *denaturalize* familiar texts, and Goldstone and Underwood [8] show that topics can surface gradual transformations invisible to explicit debate. Both caution against treating model output as a conclusion. So do the technical diagnostics: Chang et al. [4] demonstrate that better-fitting models can be *less* interpretable, and Mimno

et al. [14] formalize semantic coherence as a guard against topics that are statistically fitted but humanly meaningless. More recent work sharpens both cautions: Hoyle et al. [10] show that automated coherence metrics can crown models that human readers reject, and Da [6] argues that many computational literary findings are statistically fragile or non-reproducible. We take these as reasons to keep human interpretation in the loop and—as Section 3.4 makes explicit—to refuse coherence as the sole arbiter of model size, rather than as grounds to abandon the method.

On the substantive side, the scholarship of masculinity supplies both the formations we look for and the historiography against which we test them. Connell [5] makes plurality analytically primary and warns against reducing masculinity to a single dictionary. Davidoff and Hall [7] and Tosh [23] trace how middle-class manhood was produced through family, business, and religion, with domesticity a central but contradictory ideal. Sussman [21], Adams [1], Solinger [20], and Mallett [13] track masculine styles across gentility, asceticism, muscular Christianity, aestheticism, and self-fashioning. This body of work both motivates a broad operational vocabulary and provides the external benchmark—an expected chronology of domestic, adventurous, and commercial manhood—against which a model’s trajectories can be judged.

3 Data and methods

3.1 Corpus and segmentation

The corpus is the English-language subset of Andrew Piper’s txtLAB Novel450 dataset [16]: 150 British and American novels published between 1771 and 1930. Each novel is divided into non-overlapping, sentence-aware windows of approximately 2,000 words, yielding 9,271 segments and 6,562,588 retained tokens (mean $\approx$ 708 tokens per segment after preprocessing). Segmentation makes the model sensitive to local thematic variation—a single novel may contain domestic, commercial, and adventurous passages—while non-overlapping windows limit dependence between adjacent units. Because segments from the same novel nevertheless remain correlated, all prevalence estimates are aggregated to the novel level before comparison (Section 3.5). The 150 novels span five periods—Georgian (41), Early Victorian (40), Late Victorian (36), Edwardian (18), and Early Modernist (15)—and two author-gender categories (81 male, 69 female).

3.2 Preprocessing

Preprocessing is deliberately conservative: tokens are lowercased and normalized, common stop-words and dialogue contractions removed, conservative plural normalization applied, and very rare or ubiquitous terms excluded (retained in at least 25 and at most 65% of documents). A document-specificity filter removes 2,604 terms concentrated in individual novels, reducing the chance that protagonist names or single-work vocabulary become spurious topics; the final vocabulary holds 15,946 terms. Masculinity-relevant common words (e.g. *sir*, *gentleman*, *father*, *master*, *man*, *men*, *work*, *money*) are protected from removal. To verify that this protection does not drive the results, we re-ran the entire pipeline at $K = 7$ with no protected terms, so that all vocabulary was subject to the same frequency and document-specificity filters. The vocabulary changed by a single term (15,945 vs. 15,946), the same six formations emerged, and every period trajectory was unchanged within rounding (for example, professional breadwinning $0.02 \rightarrow 0.39$ vs. $0.02 \rightarrow 0.37$; sentimental manhood $0.35 \rightarrow 0.02$ in both). The protected list is therefore a convenience for interpretation, not a determinant of the findings; more generally, because the masculine families are applied only after estimation (Section 4.3), they shape labels rather than the topics, prevalences, or trajectories the model produces.

3.3 Model and theory-guided labeling

Models are estimated with the `stm` package [19] using Spectral initialization and the prevalence formula `period + author_gender`. Interpretation is guided by the seven masculine families introduced above, derived a priori from the scholarship in Section 2. A topic is flagged as a candidate for a family when at least two of its highest-probability or FREX (frequency–exclusivity) words fall in that family’s vocabulary; final labels reflect a reading of probability words, FREX words, and representative segments. Crucially, the families are an interpretive lens applied *after* estimation; they never enter model fitting or the choice of model size.

3.4 Choosing the number of topics

We compared models from $K = 3$ to $K = 15$ on held-out likelihood, residuals, semantic coherence, and exclusivity (Table 1, Fig. 2). Three diagnostics—held-out likelihood, residuals, and exclusivity—improve monotonically with K , as they generally do; semantic coherence is highest at low K (-27.0 at $K = 3$, -30.3 at $K = 5$) and falls thereafter. The diagnostics thus disagree, and coherence alone would favor a small model. We therefore fix K on a substantive criterion motivated by modernization theory rather than by model fit: $K = 7$ is the smallest model in which the professional-commercial formation—the central historical object of the study—separates as its own topic (it is absent at $K = 5$ and $K = 6$; Table 3), with no family duplicated. We note candidly that this criterion privileges one theoretically central formation and is thus not wholly independent of the family scheme; what it avoids is the stronger circularity of choosing K to maximize the *number* of families recovered, which would make overall recovery tautological. Models at $K = 5, 6, 8$, and 9 are retained as robustness checks, and the central historical pattern does not hinge on $K = 7$. In every neighbouring model in which the relevant topic exists, the professional-commercial formation rises across the period (Georgian to Early Modernist: $0.02 \rightarrow 0.30$ at $K = 8$, $0.01 \rightarrow 0.24$ at $K = 9$) and the sentimental and gentlemanly formations decline; topic boundaries become more compressed at lower K and more fragmented at higher K , but the direction of the trajectories is stable.

Table 1: Model comparison across candidate topic numbers ($K = 3$ – 15). Held-out likelihood, residuals, and exclusivity improve monotonically with K ; semantic coherence is best at low K and declines thereafter, so the diagnostics do not identify a single optimum.

K	Held-out	Residual	Sem. coherence	Exclusivity
3	−8.343	2.541	−26.99	8.10
4	−8.321	2.476	−30.57	8.49
5	−8.300	2.428	−30.27	8.60
6	−8.287	2.395	−35.06	8.71
7	−8.277	2.371	−39.70	8.93
8	−8.267	2.329	−40.82	9.03
9	−8.256	2.313	−41.26	9.12
10	−8.250	2.283	−41.34	9.14
11	−8.244	2.259	−46.58	9.18
12	−8.238	2.246	−42.10	9.21
13	−8.231	2.228	−44.87	9.23
14	−8.225	2.214	−44.90	9.30
15	−8.222	2.194	−43.35	9.31

3.5 Estimating uncertainty

Because the 9,271 segments are not independent—many share a novel—we work at the *novel level*: segment proportions are averaged within each novel. For each formation we then report two quantities. Per-group 95% confidence intervals (Figs. 3, 5) come from bootstrap resampling of novels within each group. For hypothesis testing we bootstrap the *between-group difference* directly—the change between the first and last periods, and the male–female contrast—over 10,000 resamples, reporting its 95% interval and a two-sided bootstrap p -value. We test the difference rather than inspect whether two per-group intervals overlap, because non-overlap of separate intervals is an unduly conservative and unreliable proxy for a difference test. All resampling is of whole novels, reflecting the true number of independent observations. Because twelve primary contrasts are reported (six period changes and six gender contrasts), we additionally apply a Benjamini–Hochberg false-discovery-rate correction; every contrast remains significant (largest adjusted $p = 0.015$, for the gentlemanly-status decline), so the substantive conclusions are unchanged. Segment-level effect estimates from `estimateEffect` were also computed; because they treat correlated segments as independent they yield implausibly extreme p -values and are not used for significance, illustrating precisely the false-precision hazard noted in Section 1.

4 Results

4.1 Formations and topic quality (RQ1)

The seven-topic model is summarized in Table 2. Six topics map onto six theoretical families; a seventh (T3) matches no masculine family and is read as a non-masculine Gothic-romance formation (*emily, chamber, castle, convent, marquis*), retained as an interpretive control that absorbs Georgian Gothic vocabulary which would otherwise contaminate the masculine topics. On topic-quality diagnostics (Fig. 1), gentlemanly status (T1) and sentimental manhood (T5) are the most coherent and distinctive; the Gothic control (T3) and aristocratic chivalry (T6) are the least coherent.

Two labels require explicit caution, because the model’s words run ahead of the category name. Imperial adventure (T2) is lexically an Anglo-American wilderness-and-maritime formation (*ship, sea, wolf, snow*; signatures of London, Melville, and Hardy) rather than a narrowly imperial one; its imperial resonance is a generic and scholarly association, not a lexical fact. Sentimental manhood (T5) is an epistolary affective register (*affection, esteem, sentiment, madam*) not specific to male characters. We retain these as formation headings while flagging that each topic is broader than its name—an instance of the interpretive discipline the method demands.

Table 2: The seven-topic model: candidate label, leading FREX words, and topic quality (semantic coherence and exclusivity; higher is better). T3 (Gothic Romance) is a non-masculine contextual control.

Topic	Candidate label	Leading FREX words	Coh.	Excl.
T1	Gentlemanly Status	miss, lady, elizabeth, fanny, mrs, lord	−22.8	9.59
T5	Sentimental Manhood	affection, conduct, esteem, madam, sentiment	−28.3	8.41
T2	Imperial Adventure	boat, fang, ship, deck, buck, wolf	−37.1	8.81
T4	Professional Breadwinner	tom, joe, dollar, cent, office, railroad	−37.1	8.99
T7	Domestic Paternalism	margaret, laura, mary, ellen, loved, mother	−37.6	9.36
T3	Gothic Romance (<i>control</i>)	emily, chamber, castle, convent, marquis	−48.1	9.41
T6	Aristocratic Chivalry	prince, knight, nation, royal, spain	−48.5	8.37

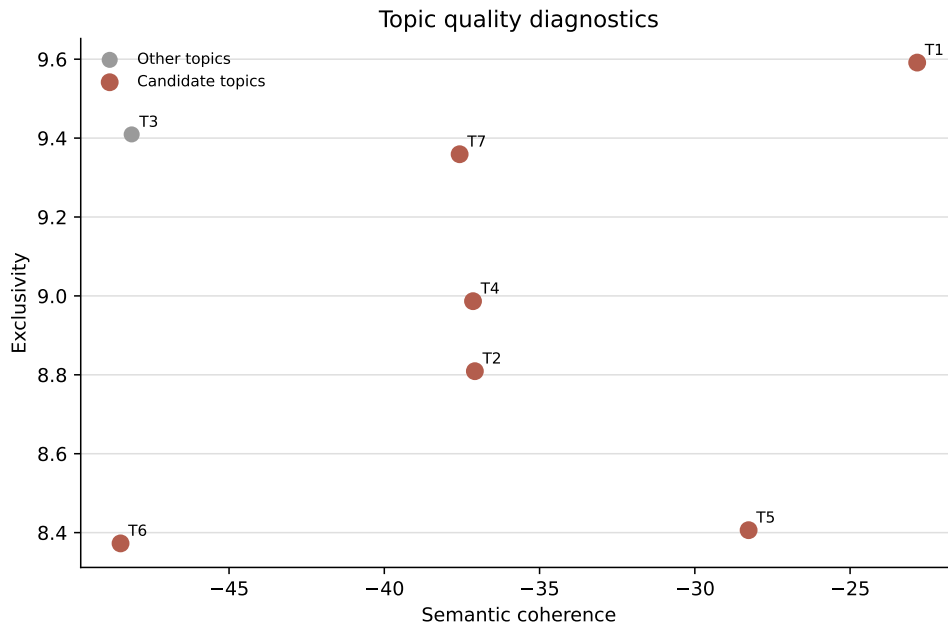


Figure 1: Topic-quality diagnostics (semantic coherence vs. exclusivity). Candidate masculine formations in colour; the Gothic-romance control in grey.

4.2 Which formations separate, and when

A second, more revealing result concerns not the seven-topic solution alone but the order in which the formations emerge as the model grows (Table 3, Fig. 2). Four families—domestic paternalism, gentlemanly status, imperial adventure, sentimental manhood—separate by $K = 5$; aristocratic chivalry separates at $K = 6$; professional breadwinning at $K = 7$; and Christian moralism not until $K = 9$. Even then it does not form a tidy devotional topic: at $K = 9$ the religious vocabulary (*religion*, *christian*, *divine*) emerges blended with philosophical and national-historical language (*philosopher*, *nation*, *government*, with exemplars such as *Hypatia* and *Romola*), and its appearance fragments the domestic topic. We read this not as the absence of religious masculinity but as evidence of *dispersion*: religious vocabulary is spread across domestic-household and historical-intellectual registers rather than concentrated, so at $K = 7$ it has no distinct topic of its own. In this corpus, moral-religious manhood is articulated through other formations rather than as a freestanding one—a pattern consistent with accounts of Victorian domestic and moral manliness [21, 23]. All seven families are retained as the theoretical lens; their differential separability is itself a finding about the corpus, not a defect of the model.

4.3 Historical trajectories (RQ2)

Figure 3 shows novel-level prevalence by period with 95% bootstrap intervals; Figure 4 gives the same values as a heatmap. Every between-period change (Early Modernist minus Georgian) is statistically robust under the difference bootstrap, but the changes differ enormously in size. Two signals dominate: professional breadwinning rises by $+0.351$ [$+0.264, +0.441$] ($p < 0.001$; from 0.021 to 0.372), and sentimental manhood falls by -0.332 [$-0.399, -0.266$] ($p < 0.001$; from 0.352 to 0.020). Domestic paternalism rises ($+0.147$ [$+0.096, +0.202$], $p < 0.001$; peaking in the Early and Late Victorian periods at 0.244 and 0.249) and imperial adventure climbs to a Late Victorian

Table 3: Family coverage across K . All seven theory-driven families are retained as the interpretive lens; the column gives the smallest K at which each separates as a distinct topic. The Gothic-romance topic is a non-masculine control and is not a family.

Family	Example keywords	First K recovered	Status at $K = 7$
Domestic Paternalism	child, family, father, mother	5	Separate
Gentlemanly Status	conduct, dignity, gentleman, sir	5	Separate
Imperial Adventure	army, captain, sea, ship	5	Separate
Sentimental Manhood	affection, feeling, grief, love	5	Separate
Aristocratic Chivalry	castle, chivalry, knight, prince	6	Separate
Professional Breadwinner	bank, business, money, trade	7	Separate
Christian Moralism	christian, church, faith, soul	9	Fused with Domestic

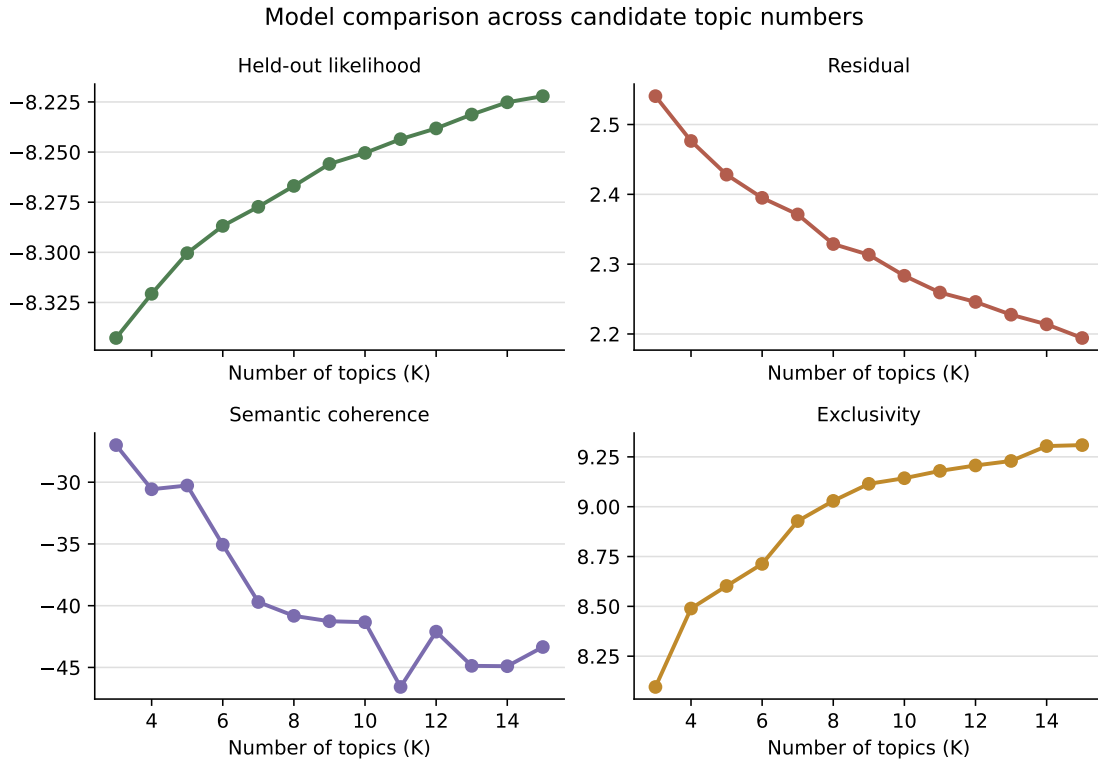


Figure 2: Model comparison for $K = 3$ –15. Held-out likelihood, residuals, and exclusivity improve monotonically with K , while semantic coherence peaks at low K and declines. The choice of $K = 7$ rests on the substantive criterion in Section 3.4, not on these curves alone.

plateau (+0.150 [+0.071, +0.229], $p < 0.001$). The Gothic control falls steeply (0.188 $\rightarrow$ 0.036), confirming that the model tracks broad literary-historical structure rather than gendered vocabulary alone.

The two “rank” formations also decline significantly, but modestly. Gentlemanly status falls by -0.074 [$-0.136, -0.016$] ($p = 0.014$; from a 0.198 Early Victorian peak to 0.077) and aristocratic chivalry by -0.090 [$-0.154, -0.026$] ($p = 0.004$). The “rank-to-work” narrative is therefore asymmetric not in *significance* but in *magnitude*: the rise of professional-commercial masculinity is roughly five times larger than the erosion of gentlemanly recognition. A reading on which gentility is simply swapped for work overstates the second half of the exchange; the data show a large new formation emerging *alongside* a real but modest decline of the old. (Inspecting per-group interval overlap, by contrast, would have wrongly labeled the gentlemanly decline “non-significant”—an artifact of that heuristic, not of the data.)

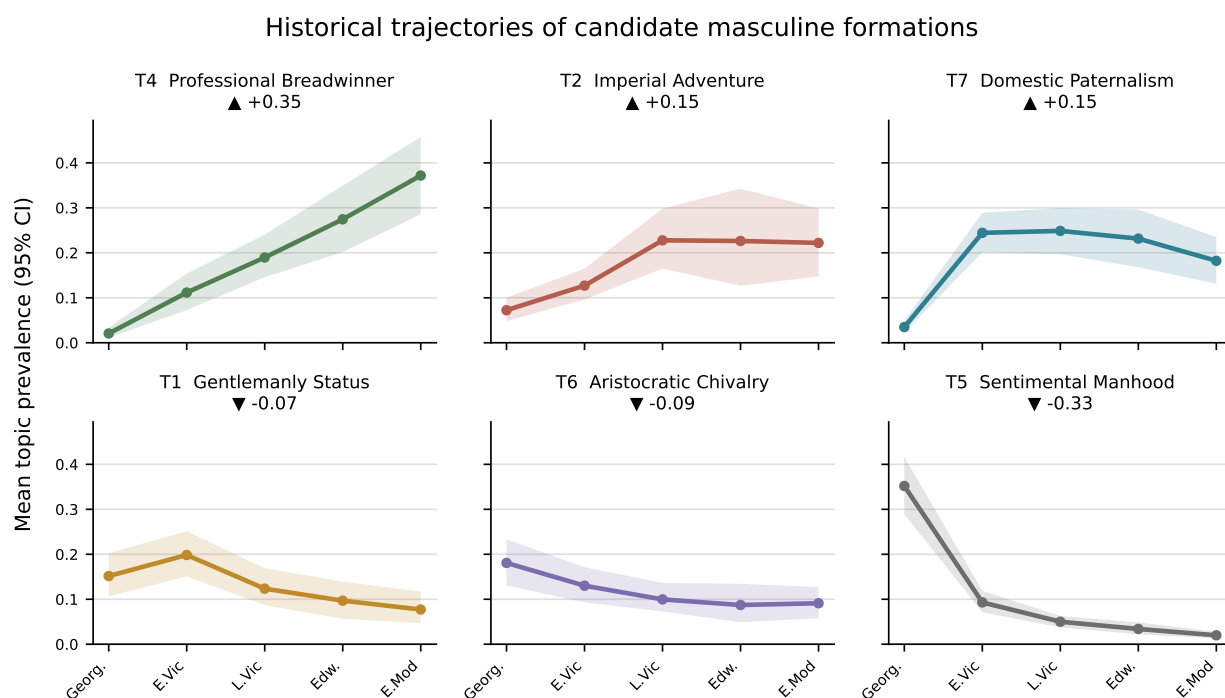


Figure 3: Historical trajectories of the six candidate masculine formations (novel-level mean prevalence with 95% bootstrap intervals), ordered by net change. Rising: professional breadwinning, imperial adventure, domestic paternalism. Declining: gentlemanly status, aristocratic chivalry, sentimental manhood.

4.4 Author gender (RQ3)

Figure 5 and Table 4 report novel-level prevalence by author gender. Under the difference bootstrap, all six male–female contrasts are statistically robust: female authors are higher on gentlemanly status (-0.087 male–female, $p < 0.001$), domestic paternalism (-0.089 , $p < 0.001$), and sentimental manhood (-0.100 , $p = 0.001$), whereas male authors are higher on imperial adventure ($+0.103$, $p < 0.001$), aristocratic chivalry ($+0.107$, $p < 0.001$, the largest gap), and professional breadwinning ($+0.074$, $p = 0.005$). The professional-breadwinning gap is worth a note: the per-group intervals in Fig. 5 nearly touch, yet the paired difference is robust—another instance in which interval overlap

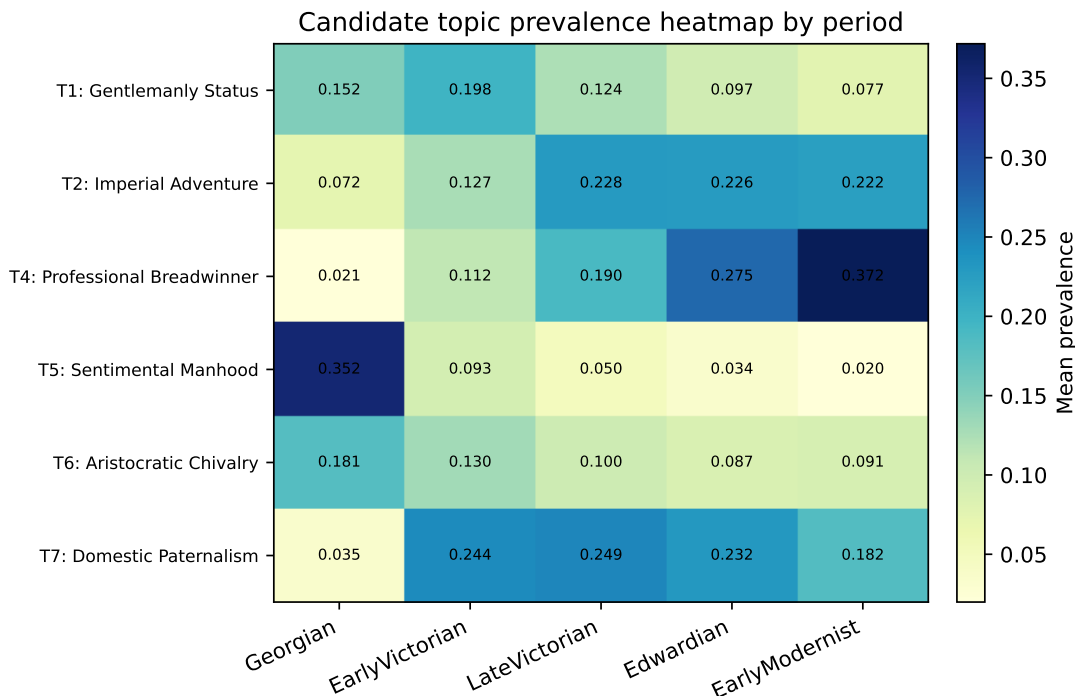


Figure 4: Heatmap of novel-level mean prevalence by period for the six candidate formations.

understates a real difference. Because author gender enters only as a secondary covariate, we make no causal claim about gendered authorship; the contrasts describe association, not authorial intent.

Table 4: Novel-level mean prevalence by author gender, with the bootstrapped male–female difference and its two-sided p -value (candidate formations). All six contrasts are statistically robust.

Formation	Female	Male	M–F diff [95% CI]	p
Gentlemanly Status	0.191	0.103	−0.087 [−0.132, −0.043]	< 0.001
Domestic Paternalism	0.229	0.139	−0.089 [−0.139, −0.042]	< 0.001
Sentimental Manhood	0.193	0.093	−0.100 [−0.158, −0.042]	0.001
Imperial Adventure	0.102	0.205	+0.103 [+0.054, +0.153]	< 0.001
Aristocratic Chivalry	0.070	0.177	+0.107 [+0.070, +0.143]	< 0.001
Professional Breadwinner	0.111	0.185	+0.074 [+0.023, +0.125]	0.005

5 Discussion

The substantive result is a reweighting, not a replacement. Across the corpus, fiction shifts from masculine formations organized around gentility, rank, and sentiment toward one organized around work, money, and the city, while the older formations persist at lower prevalence rather than disappearing. This is a stronger claim than a word-frequency trend, because it rests on topics—stable constellations of co-occurring words—yet a weaker claim than an account of masculine character, because the method models lexical environments rather than narrative agency.

Three features of the results deserve emphasis. First, the trajectories show *convergent validity* with independent historiography: domestic paternalism peaks in the high-Victorian decades

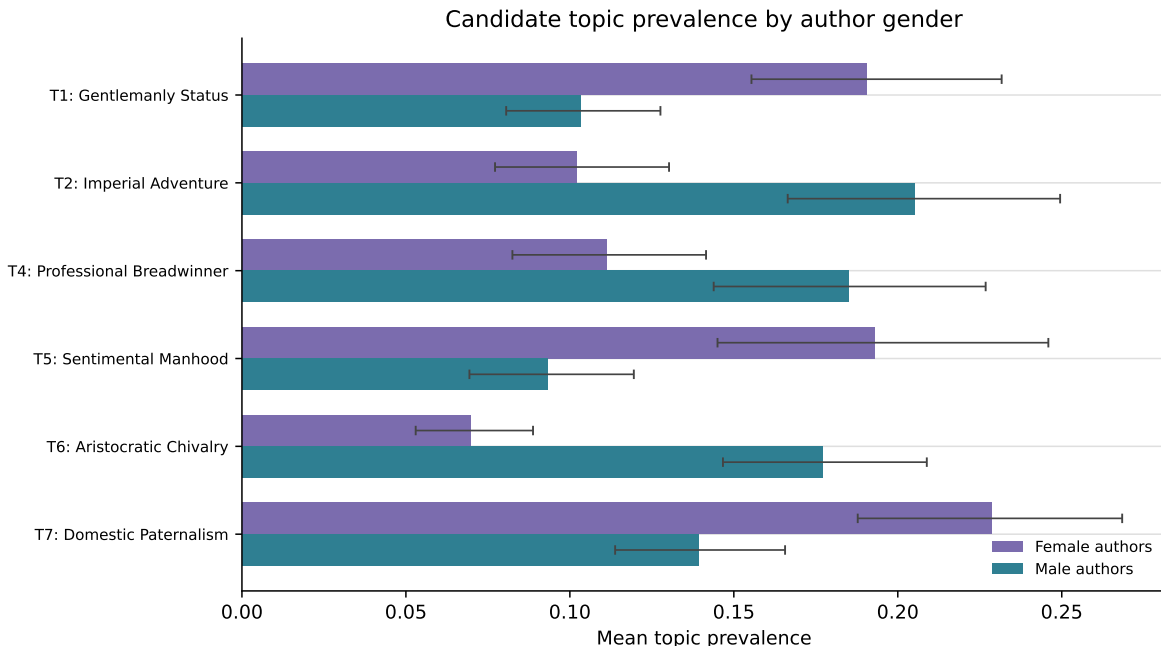


Figure 5: Novel-level mean prevalence by author gender with 95% bootstrap error bars (per-group). All six male–female differences are robust under the difference bootstrap (Table 4); for professional breadwinning the per-group bars nearly overlap even though the paired difference is significant ($p = 0.005$).

[23], imperial adventure crests in the late century, and the professional-commercial formation rises into modernity as status is reorganized from inherited rank toward labor and capital. That an unsupervised model recovers a periodization scholars established by close reading *corroborates* the formations; it does not prove them, since the labels remain interpretive and period effects may partly reflect genre and corpus composition (for instance, the late-century concentration of adventure and urban fiction). Second, the model supplies a mild corrective: although the “decline of the gentleman” is statistically robust, it is roughly five times smaller in magnitude than the rise of the breadwinner, so the common framing of a symmetric swap from gentility to work overstates the losing side of the ledger. This asymmetry of magnitude is easy to miss without an explicit difference test, and is a concrete payoff of resampling whole novels and testing the contrast directly rather than reading significance off overlapping per-group intervals.

Third, and most general, the study clarifies what topic modeling can and cannot establish about plurality. It can show that multiple masculine formations are lexically distinguishable and that their prevalence is reweighted over time— operationalizing Connell’s plurality at corpus scale. It cannot certify *how many* formations exist: the number of topics K is a researcher choice, the diagnostics conflict on the “best” K (Section 3.4), and the families separate at different K rather than all at once. The late emergence of a standalone Christian-moralism topic is, on this reading, evidence of fusion—religious manhood expressed through domesticity—rather than evidence that the formation is absent. Treating the count of masculinities as an empirical output would reintroduce exactly the circularity we set out to avoid.

6 Limitations

Four limitations bound these claims. First, the study includes no independent human annotation, so it can say that topics are meaningfully *associated* with recognized masculine roles, not that they *are* masculinity. Second, with 150 novels, period effects may partly reflect genre distribution, author selection, and availability biases, and the smaller late-period samples produce wider intervals. Third, although novel-level aggregation addresses the dependence among same-novel segments, a fully hierarchical model would be preferable. Fourth, some topics remain contaminated by character names (e.g. *tom*, *emily*, *margaret*) despite the document-specificity filter, and several labels (notably imperial adventure and sentimental manhood) are broader than their names; both are reasons to read the formations as maps rather than as definitions.

7 Conclusion

Using structural topic modeling, this article maps how masculine social formations are reweighted across English fiction from 1771 to 1930. Six of seven theory-driven formations separate as distinct topics; the seventh, Christian moralism, remains fused with domesticity until a larger model size. Novel-level bootstrap intervals show a robust rise of professional-commercial masculinity and a robust decline of a sentimental register, alongside a markedly weaker decline of gentlemanly recognition. Beyond this case, the study offers a disciplined template for applying topic models to theoretical constructs: treat the categories as a lens rather than a target, fix model size on a substantive criterion, quantify uncertainty at the level of genuinely independent observations, and validate against external scholarship. On that basis, computational analysis can make the plurality and historical movement of masculine formations measurable—while leaving the question of their number where it belongs, with theory and interpretation rather than with the model.

Data and code availability

The custom code central to this study—segmentation, STM preparation, model estimation, reporting, model comparison, family-coverage analysis, and figure generation (`proposal2_segment.py`, `proposal2_stm_prep.py`, `proposal2_run_stm.R`, `proposal2_stm_report.py`, `proposal2_family_coverage.py`, `proposal2_hssc_figures.py`)—together with the derived segment metadata, STM input matrices, topic-word tables, model-comparison outputs, bootstrap summaries, and figure scripts, are available in an anonymized repository for review at [URL] and will be deposited in a persistent, DOI-providing archive (OSF/Zenodo/figshare) on acceptance. The underlying corpus is openly available as Andrew Piper’s txtLAB Novel450 dataset [16].

Ethics declarations

Ethical approval was not required: the study analyses published literary texts and involves no human participants or personal data. The authors declare no competing interests.

A Theory-driven family keyword lists

The seven masculine families are an a-priori interpretive lens (Section 4.3). A topic is flagged as a candidate for a family when at least two of its highest-probability or FREX words fall in that

family’s list. The lists below are the exact vocabularies used; they are applied only *after* estimation and never enter model fitting or the choice of K . The Gothic-romance topic matches no family and is treated as a non-masculine contextual control.

Gentlemanly Status: address, character, conduct, courtesy, dignity, gentleman, gentlemen, honor, honour, lady, ladyship, polite, propriety, rank, reputation, respect, respectable, sir, society.

Domestic Paternalism: boy, care, child, children, daughter, family, father, girl, home, household, husband, marriage, married, mother, parent, son, wife.

Aristocratic Chivalry: baron, castle, chivalry, court, crown, duke, hero, highness, honor, honour, horse, king, knight, lord, master, noble, prince, queen, royal, sword.

Professional Breadwinner: bank, business, capital, clerk, credit, debt, dollar, employment, fortune, income, labor, labour, merchant, money, office, pound, profession, property, trade, work.

Imperial Adventure: army, battle, camp, captain, colonial, colony, courage, danger, empire, endurance, enemy, fight, forest, frontier, indian, island, jungle, native, officer, rifle, scout, sea, ship, soldier, travel, voyage, war, wilderness.

Christian Moralism: christian, church, conscience, duty, evil, faith, god, goodness, heaven, holy, minister, moral, piety, prayer, priest, religion, sin, soul, truth, virtue.

Sentimental Manhood: affection, beloved, dear, emotion, feeling, grief, happiness, joy, love, melancholy, misery, passion, sensibility, sentiment, sorrow, sympathy, tender.

References

- [1] James Eli Adams. *Dandies and Desert Saints: Styles of Victorian Masculinity*. Cornell University Press, Ithaca, NY, 1995.
- [2] David M. Blei, Andrew Y. Ng, and Michael I. Jordan. Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022, 2003.
- [3] Rachel Sagner Buurma. The fictionality of topic modeling: Machine reading Anthony Trollope’s Barsetshire series. *Big Data & Society*, 2(2):1–6, 2015. doi: 10.1177/2053951715610591.
- [4] Jonathan Chang, Sean Gerrish, Chong Wang, Jordan L. Boyd-Graber, and David M. Blei. Reading tea leaves: How humans interpret topic models. In *Advances in Neural Information Processing Systems 22*, 2009.
- [5] R. W. Connell. *Masculinities*. University of California Press, Berkeley, CA, 1995.
- [6] Nan Z. Da. The computational case against computational literary studies. *Critical Inquiry*, 45(3):601–639, 2019. doi: 10.1086/702594.
- [7] Leonore Davidoff and Catherine Hall. *Family Fortunes: Men and Women of the English Middle Class, 1780–1850*. University of Chicago Press, Chicago, IL, 1987.
- [8] Andrew Goldstone and Ted Underwood. The quiet transformations of literary studies: What thirteen thousand scholars could tell us. *New Literary History*, 45(3):359–384, 2014. doi: 10.1353/nlh.2014.0025.
- [9] Ryan Heuser and Long Le-Khac. A quantitative literary history of 2,958 nineteenth-century british novels: The semantic cohort method. Pamphlet 4, Stanford Literary Lab, 2012.

- [10] Alexander Hoyle, Pranav Goel, Andrew Hian-Cheong, Denis Peskov, Jordan Boyd-Graber, and Philip Resnik. Is automated topic model evaluation broken? the incoherence of coherence. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.
- [11] Matthew L. Jockers. *Macroanalysis: Digital Methods and Literary History*. University of Illinois Press, Urbana, IL, 2013.
- [12] Matthew L. Jockers and David Mimno. Significant themes in 19th-century literature. *Poetics*, 41(6):750–769, 2013. doi: 10.1016/j.poetic.2013.08.005.
- [13] Phillip Mallett, editor. *The Victorian Novel and Masculinity*. Palgrave Macmillan, London, UK, 2015. doi: 10.1057/9781137491541.
- [14] David Mimno, Hanna M. Wallach, Edmund Talley, Miriam Leenders, and Andrew McCallum. Optimizing semantic coherence in topic models. In *Proceedings of the 2011 Conference on Empirical Methods in Natural Language Processing*, pages 262–272, Edinburgh, UK, 2011.
- [15] Franco Moretti. Conjectures on world literature. *New Left Review*, (1):54–68, 2000.
- [16] Andrew Piper. txtLAB Novel450: A multilingual data set of novels, 2016. Dataset.
- [17] Andrew Piper. *Enumerations: Data and Literary Study*. University of Chicago Press, Chicago, IL, 2018.
- [18] Margaret E. Roberts, Brandon M. Stewart, Dustin Tingley, Christopher Lucas, Jetson Leder-Luis, Shana Kushner Gadarian, Bethany Albertson, and David G. Rand. Structural topic models for open-ended survey responses. *American Journal of Political Science*, 58(4):1064–1082, 2014. doi: 10.1111/ajps.12103.
- [19] Margaret E. Roberts, Brandon M. Stewart, and Dustin Tingley. stm: An R package for structural topic models. *Journal of Statistical Software*, 91(2):1–40, 2019. doi: 10.18637/jss.v091.i02.
- [20] Jason D. Solinger. *Becoming the Gentleman: British Literature and the Invention of Modern Masculinity, 1660–1815*. Palgrave Macmillan, New York, NY, 2012. doi: 10.1057/9780230391840.
- [21] Herbert Sussman. *Victorian Masculinities: Manhood and Masculine Poetics in Early Victorian Literature and Art*. Cambridge University Press, Cambridge, UK, 1995.
- [22] Timothy R. Tangherlini and Peter Leonard. Trawling in the sea of the great unread: Sub-corpus topic modeling and humanities research. *Poetics*, 41(6), 2013.
- [23] John Tosh. *A Man’s Place: Masculinity and the Middle-Class Home in Victorian England*. Yale University Press, New Haven, CT, 1999.
- [24] Ted Underwood. *Distant Horizons: Digital Evidence and Literary Change*. University of Chicago Press, Chicago, IL, 2019.
- [25] Ted Underwood, David Bamman, and Sabrina Lee. The transformation of gender in English-language fiction. *Journal of Cultural Analytics*, 2018. doi: 10.22148/16.019.